Doubling Dimension and Treewidth

Felix Zhou*

Abstract

The dimension of a normed linear space is defined as the size of a maximally linearly independent set. For general metric spaces, the *doubling dimension* has been proposed as a generalization of dimension. In this extended survey, we present an embedding of metric spaces with low doubling dimension into the so-called graphs of bounded treewidth due to Talwar [Tal04]. This implies approximation algorithms for many distance-based problems such as the Travelling Salesman Problem.

Contents

1 Introduction **2**
1.1 Previous Work . . . 2
1.2 Techniques . . . 2

2 Background **3**
2.1 Metric Spaces . . . 3
2.2 Doubling Dimension . . . 4
2.3 Embeddings . . . 7
2.4 Treewidth & Tree Decompositions . . . 7

3 The Embedding **9**
3.1 The Split-Tree . . . 9
3.1.1 Nested Covers . . . 10
3.1.2 Split-Tree Construction . . . 11
3.1.3 Analysis . . . 11
3.2 The Algorithm . . . 15
3.3 Analysis . . . 16
3.3.1 Expected Distortion . . . 16
3.3.2 Treewidth . . . 17

4 Approximation **18**
4.1 Bounding the Aspect Ratio . . . 19
4.2 Obtaining a Solution in the Net . . . 20
4.2.1 Monadic Second-Order Logic . . . 20
4.2.2 Dynamic Programming . . . 21
4.2.3 Probability & Linear Algebra . . . 21
4.3 Obtaining an Approximate Solution . . . 21
4.4 Repeated Trials . . . 22
4.5 Other Problems . . . 23

*David R. Cheriton School of Computer Science, University of Waterloo, Waterloo, Ontario N2L 1A2, Canada. *cfzhou@uwaterloo.ca*

5 Miscellaneous Results **23**
5.1 A PTAS for TSP . . . 23
5.2 Connections to the Cantor Set . . . 24
5.2.1 The Cantor Set . . . 24
5.2.2 Compactness . . . 25
5.2.3 Total Disconnectedness . . . 25
5.2.4 A Dead-End . . . 26

6 Conclusion **26**

7 References **27**

8 Appendix **28**

1 Introduction

Metric spaces capture the notion of distance. From an analyst's point of view, our intuition for Euclidean spaces such as $\mathbb{R}^k$ transfer quite easily to metric spaces. Thus it is not too difficult to translate the theory for simpler spaces to abstract metric spaces. On the other hand, the rich algorithmic techniques for Euclidean spaces rely more so on the geometry of these spaces. Therefore, it tends to be more difficult to adapt the exact algorithms for $\mathbb{R}^k$ to abstract metric spaces.

The doubling dimension of a metric space roughly speaking puts a growth restriction on the geometry of the space. Such notions arise naturally in practical applications such as peer-to-peer networks [NH02] and data analysis [TSL00].

In this extended survey, we examine in detail a quasi-polynomial approximation scheme (QPTAS) for the Traveling Salesman Problem on metric spaces of bounded doubling dimension due to Talwar [Tal04]. This is significant since (not necessarily metric) TSP does not in general admit even constant approximations. The celebrated work of Arora [Aro98] shows that in the restricted case of Euclidean spaces with constant dimension, a polynomial time approximation scheme (PTAS) can be achieved. Talwar's work somewhat relaxes the Euclidean restriction.

1.1 Previous Work

The definition of doubling dimension was due to Gupta et al. [GKL03] loosely inspired by the work of Assouad [Ass83].

Approximation schemes for numerous problems in Euclidean spaces have been extensively studied. See [Aro03] for a survey.

Metric embedding is a rich field of study on its own. The book by the late Matousek [Mat13] is an excellent resource.

The treewidth of a graph has been explored from multiple directions. See [SS10] for the closely related theory of "pursuit-evasion games". [Bie20] approaches treewidth from the angle of chordal graphs and explains in detail the powerful dynamic programming techniques afforded to us by the tree decomposition.

1.2 Techniques

Talwar's result leverages a randomized data structure called the split-tree serving a similar purpose to the quadtree for $\mathbb{R}^2$. We will reorganize Talwar's scheme and re-interpret it in terms of metric embeddings and treewidth, two powerful tools of algorithm design. While the interpretation loses some of the cleverness of Talwar's scheme, it provides a generalized framework for approximating other distance-related problems.

The arguments used in Talwar's paper follows closely from [Fak+03]. The version we give is partially expanded from [Tal04] and partially inspired by [Fel+18].

2 Background

We first provide some definitions and show some simple properties.

2.1 Metric Spaces

Let us introduce some elementary analysis.

Definition 2.1 (Metric Space): A *metric* space is a pair (X, d) where X is an arbitrary set and $d : X \to [0, \infty)$ is a function satisfying the following properties:

(i) $d(x, y) = 0 \iff x = y$ for all $x, y \in X$ (positive definiteness)

(ii) $d(x, y) = d(y, x)$ for all $x, y \in X$ (symmetry)

(iii) $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$ (triangle inequality)

The canonical example of a metric space is $\mathbb{R}^n$ under the Euclidean distance. In fact, any norm $\|\cdot\|$ induces a metric where

$$d(x, y) = \|x - y\|.$$

Let $G = (V, E)$ be an undirected, positive edge-weighted graph. Let $\mathrm{dist}(u, v)$ denote the length of the shortest path between $u, v \in V$. The *shortest path metric* space induced by G is the metric space (V, dist). Observe that any finite metric space can be expressed as the shortest path metric space of a weighted complete graph where the weight of an edge is given by the distance between the endpoints.

Definition 2.2 (Ball): The *(closed) ball* of radius r around a point $x \in X$ is

$$B_r(x) := \{y \in X : d(y, x) \leq r\}.$$

See Figure 1. We say a subset $Y \subseteq X$ is *bounded* if it is contained within a ball of finite radius.

Definition 2.3 (δ-Net): Let $Y \subseteq X$ and $\delta > 0$. A *δ-cover* of Y is a subset $Z \subseteq Y$ such that

$$Y \subseteq \bigcup_{z \in Z} B_\delta(z).$$

If in addition, Z is finite, and $d(z, z') > \delta$ for all $z \neq z' \in Z$, we say Z is a *δ-net* of Y.

Remark that in the case where Y is finite, we can always build a δ-net of Y greedily. Indeed, continually add uncovered points to our net until we cover all of Y. See Figure 2 for an example of such a greedily constructed net.

Definition 2.4 (Aspect Ratio): Let $Y \subseteq X$ be finite. The aspect ratio of Y is

$$\Delta := \frac{\max_{y_1 \neq y_2 \in Y} d(y_1, y_2)}{\min_{y_3 \neq y_4 \in Y} d(y_3, y_4)}.$$

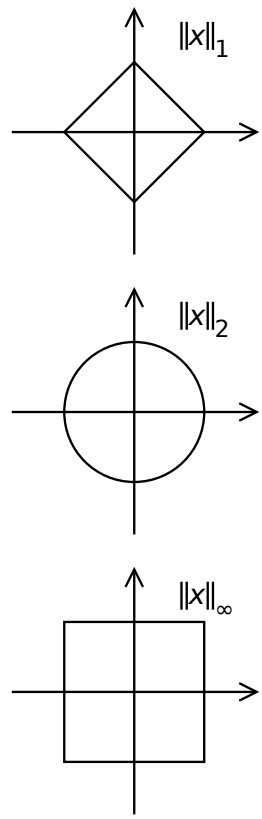

Figure 1: The unit balls in $\mathbb{R}^2$ corresponding to the ℓ_1-norm, Euclidean distance, and ℓ_∞-norm, courtesy of [Esm06].

2.2 Doubling Dimension

Here we introduce the notion of doubling dimension.

Definition 2.5 (Doubling Dimension): Let (X, d) be a metric space. The *doubling dimension* of the metric space is the smallest constant $k \in \mathbb{R}$ such that given any $x \in X, r > 0$, we have

$$B_r(x) \subseteq \bigcup_{y \in Y} B_{r/2}(y)$$

for some $Y \subseteq X$ satisfying $|Y| \leq 2^k$.

For the remainder of this survey, all metric spaces (X, d) are finite and implicitly have doubling dimension k. We will explicitly state so otherwise.

In some sense, the doubling dimension is a measure of how fast the "volume" of a metric space is growing. To gain some intuition of doubling dimension, we now prove that $\mathbb{R}^k$ has doubling dimension $\Theta(k)$.

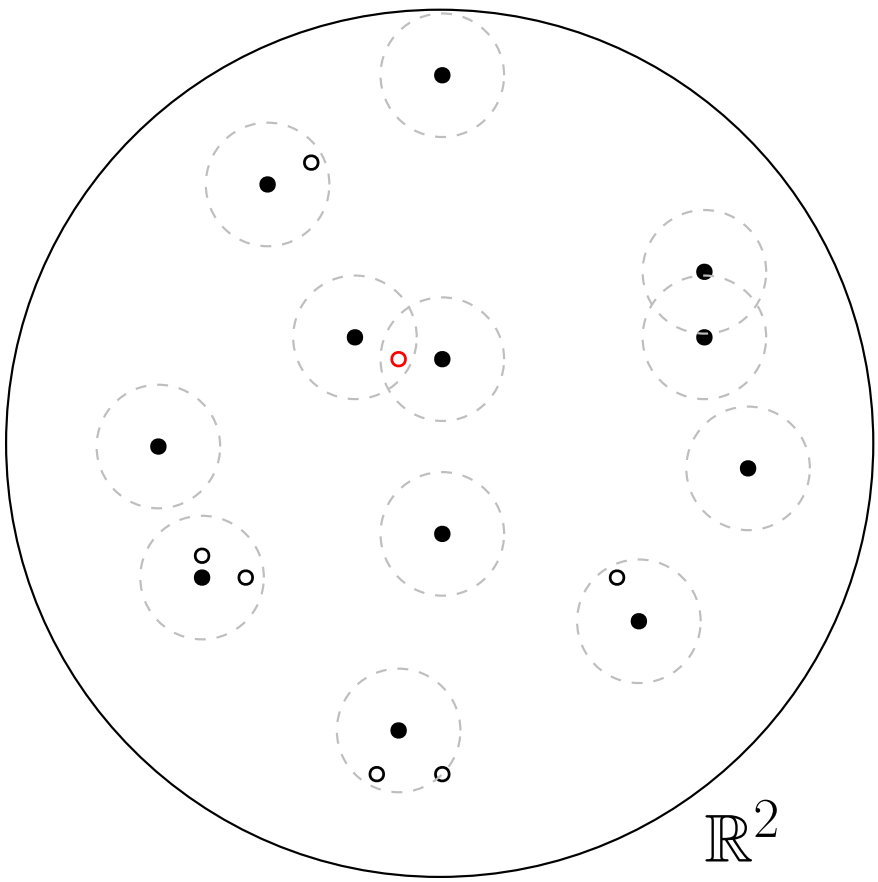

Figure 2: A greedily constructed δ-net. The solid points have been chosen as net points. Notice how choosing the red point would have covered both its neighbours so this net is not minimal.

Lemma 2.1 ([Jor14]):
The volume of a ball of radius r in $\mathbb{R}^k$ is

$$V_k(r) = \frac{r^k \pi^{k/2}}{\Gamma(k/2+1)}$$

where Γ denotes the gamma function.

A rigorous proof of this fact requires a precise definition of volume using machinery such as the Lebesgue integral as well as some non-trivial analysis and computations. We invite the reader to use their intuition regarding volume and defer the interested reader to [Jor14].

Lemma 2.2:
Every bounded subset of $\mathbb{R}^k$ has a δ-net for all $\delta > 0$.

This is a well-known fact in analysis and is not too hard to believe. The proof is essentially the Heine-Borel theorem, which equates boundedness with total boundedness in $\mathbb{R}^k$.

With Lemma 2.1 and Lemma 2.2, the following result is rather straightforwards.

Proposition 2.3:
$\mathbb{R}^k$ has doubling dimension $\Theta(k)$.

Proof (Proposition 2.3): Observe that by Lemma 2.1,

$$\frac{V_k(r)}{V_k(r/2)} = 2^k.$$

This means that the volume of a ball of radius r in $\mathbb{R}^k$ requires at least 2^k balls of radius $\frac{r}{2}$ to be covered, even if those smaller balls are disjoint. Clearly, the doubling dimension of $\mathbb{R}^k$ is $\Omega(k)$.
By Lemma 2.2, there is a $\frac{r}{2}$-net of $B_r(x)$ for all $x \in \mathbb{R}^k$. Let Z be a minimal $\frac{r}{2}$-net with respect to cardinality. Since nets are well separated by definition, the balls of radius $\frac{r}{4}$ around points of Z are disjoint. Moreover, since $Z \subseteq B_r(x)$,

$$\bigcup_{z \in Z} B_{r/2}(z) \subseteq B_{2r}(x).$$

The bound here is not tight but simplifies our computations.
Again by the volume argument, we have

$$\frac{V_k(2r)}{V_k(r/4)} = 2^{3k}.$$

This is an upper bound on $|Z|$ since the numerator upper bounds the total volume covered by Z and the denominator lower bounds how much volume each point of Z contributes. It follows that the doubling dimension of $\mathbb{R}^k$ is $O(k)$. □

The following is an important fact which comes in handy multiple times.

Proposition 2.4:
Let (X, d) be a metric space with doubling dimension k and $Y \subseteq X$ with aspect ratio Δ. Then

$$|Y| \leq 2^{k\lceil 1+\log_2 \Delta \rceil}.$$

Proof (Proposition 2.4): Let R, r be the maximum and minimum distance between points of Y respectively. Fix $y \in Y$ and notice that $Y \subseteq B_R(y)$.
But $B_R(y)$ is covered by at most 2^k balls of radius $\frac{R}{2}$, and 2^{2k} balls of radius $\frac{R}{4}$ and so on until we recurse to balls of radius at most $\frac{r}{2}$. At this point, the balls are singleton sets and we can upper bound the number of these sets instead.
We want the smallest integer ℓ such that

$$\frac{R}{2^\ell} \leq \frac{r}{2} \iff \Delta := \frac{R}{r} \leq 2^{\ell-1}.$$

Taking $\ell := \lceil 1 + \log_2 \Delta \rceil$ does the trick. It follows that

$$|Y| \leq 2^{k\lceil 1+\log_2 \Delta \rceil}.$$

□

2.3 Embeddings

Here, we give the formal definition of an embedding. A good image to keep in mind is the embedding of $\mathbb{R}^3 \to \mathbb{R}^2$ given by

$$(x, y, z) \mapsto (x, y).$$

This the example above, we lose some information about our set in exchange for a smaller description of points. The tradeoff we make in Talwar's embedding follows a similar spirit.

Definition 2.6 (Embedding): An embedding is a function $\phi : (X, d_X) \to (Y, d_Y)$.

For our purposes, it is usually clear what $\phi(x)$ is for any given x. Thus we omit the notation and simply describe Y.

Definition 2.7 (Distortion): Let (X, d_X) be finite and $\phi : (X, d_X) \to (Y, d_Y)$ an embedding. The distortion of ϕ is

$$\max_{x,x' \in X} \frac{d_Y(\phi(x), \phi(x'))}{d_X(x, x')}.$$

Note that there are alternative definitions of distortion which take into account how much points shrink within Y. However, the distance between points in an embedding can only increase for our purposes, thus the given definition suffices.

Metric embeddings are a powerful general tool for developing algorithms. If we could embed metric spaces of low doubling dimension into low dimensional Euclidean space with constant distortion, we gain access to the formidable algorithmic techniques restricted to those spaces. Unfortunately, Gupta et al. showed that this is not generally possible in [GKL03].

2.4 Treewidth & Tree Decompositions

Historically, the treewidth of a graph has been defined and discovered independently by multiple authors. For example, it can be defined in terms of the so-called "pursuit-evasion games" or the chordal completion of a graph. Most notably, Robertson and Seymour leveraged treewidth to prove the Graph Minor Theorem in a series of more than 20 papers spanning over 500 pages. Later on, treewidth gained popularity as a common parameter to capture the "hardness" of problems on graphs through fixed-parameter tractability (FPT). We will define the treewidth of a graph through tree decompositions.

Definition 2.8 (Tree Decomposition): Let $G = (V, E)$ be an undirected graph. A tree decomposition of G is a pair (T, I) where $T = (N, A)$ is a tree and $I : N \to 2^V$ assigns each node of T with a subset of the vertices of G commonly called *bags*. Furthermore,

(i) For every $v \in V$, there is some $b \in N$ such that $v \in I(b)$. We say $I(b)$ *covers* the vertex v

(ii) For every $uv \in E$, there is some $b \in N$ such that $u, v \in I(b)$. We say $I(b)$ *covers* the edge uv

(iii) For all $v \in V$, let $M_v \subseteq N$ be the nodes whose bags contain v. Then $T[M_v]$ is a connected subgraph. This is the *connectivity condition*

Notice that a tree decomposition does not dictate which edges must exist between vertices in a common bag but does forbid the existence of uncovered edges.

For the sake of clarity, we will refer to the vertex and edge sets of the original graph and the node and arc sets of the "host" tree. In addition, we tend to drop the distinction between $b, I(b)$ and simply identify the nodes with subsets of V. Thus in a slight abuse of notation, we can write $v \in b$ instead of $v \in I(b)$.

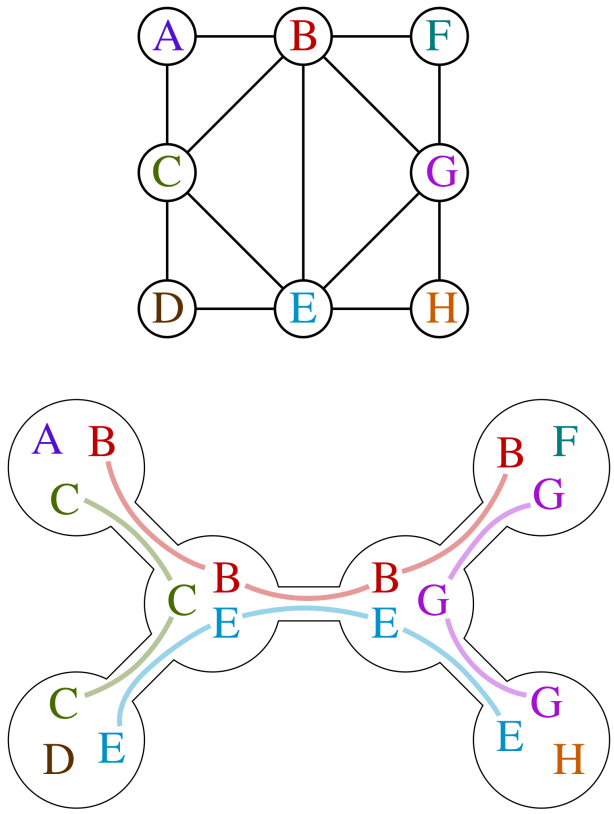

Figure 3: A graph and an accompanying tree decomposition of width 2, courtesy of [Epp10].

Every graph has a trivial tree decomposition consisting of a single bag containing all vertices. We are therefore interested in "smaller" decompositions. The *width* of a tree decomposition

$$\max_{b \in N} |b| - 1$$

is the size of the largest bag minus one. See Figure 3.

Definition 2.9 (Treewidth): The treewidth of a graph $G = (V, E)$ is the minimum width among all tree decompositions of G.

It is easy to see that a graph is a tree if and only if it has treewidth 1. Thus in some sense, treewidth can be thought of as "closeness" to a tree. Now, many NP-hard problems such as vertex cover and independent set admit exact linear-time solutions through bottom-up dynamic programming in trees. The idea behind this is that we can "join" partial solutions on subtrees to form a complete solution. This concept generalizes well to graphs of low treewidth through the tree decomposition. The motivation behind Talwar's embedding is to gain access to the well-studied algorithmic techniques around tree decompositions.

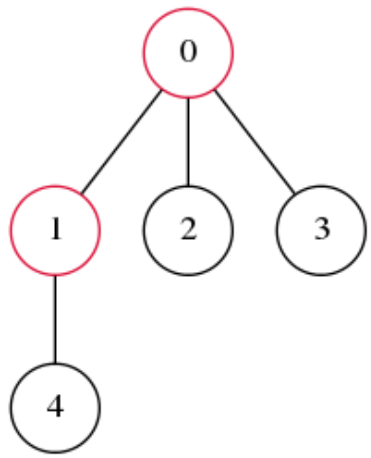

Figure 4: A minimum vertex cover in a tree.

3 The Embedding

We now present a detailed description of the embedding by Talwar. This is an independent rearrangement of [Tal04] partially done in [Fel+18]. Neither paper gives the explicit embedding in detail, which is our goal.

Theorem 3.1 ([Tal04]):
Let (X, d) be a metric space with doubling dimension k and aspect ratio Δ.
Given any $\epsilon > 0$, there is a probabilistic embedding of (X, d) into the shortest path metric space of a graph H with treewidth at most $\mathcal{K}$, for

$$\mathcal{K} \in O\left(2^k \cdot \frac{k \log_2 \Delta}{\epsilon}\right)$$

such that the expected distortion is at most $(1 + \epsilon)$.
Moreover, this embedding can be computed in polynomial time.

3.1 The Split-Tree

Recall that a quadtree for a finite subset of $\mathbb{R}^2$ is constructed by repeatedly "splitting" the points in quarters and recursing in the smaller subsets. See Figure 5. In some sense, the quadtree captures the "closeness" of points since closer points are less likely to be separated. A crucial ingredient of Talwar's embedding is the randomized *split-tree*. This data structure serves a similar purpose as the quadtree but for metrics of bounded doubling dimension.

Definition 3.1 (Partition): Let X be an arbitrary set. A partition of X is a disjoint collection of subsets, $\mathcal{P} \subseteq 2^X$, called *clusters*, such that

$$X = \bigcup_{C \subseteq \mathcal{P}} C.$$

Consider the following operation: Let $\mathcal{P}_\ell := \{X\}$ be the trivial partition of (X, d). Recursively, obtain a partition of each $C \in \mathcal{P}_i$ and let $\mathcal{P}_{i-1}$ be the union of these partitions. Continue this process until $\mathcal{P}_0$ only

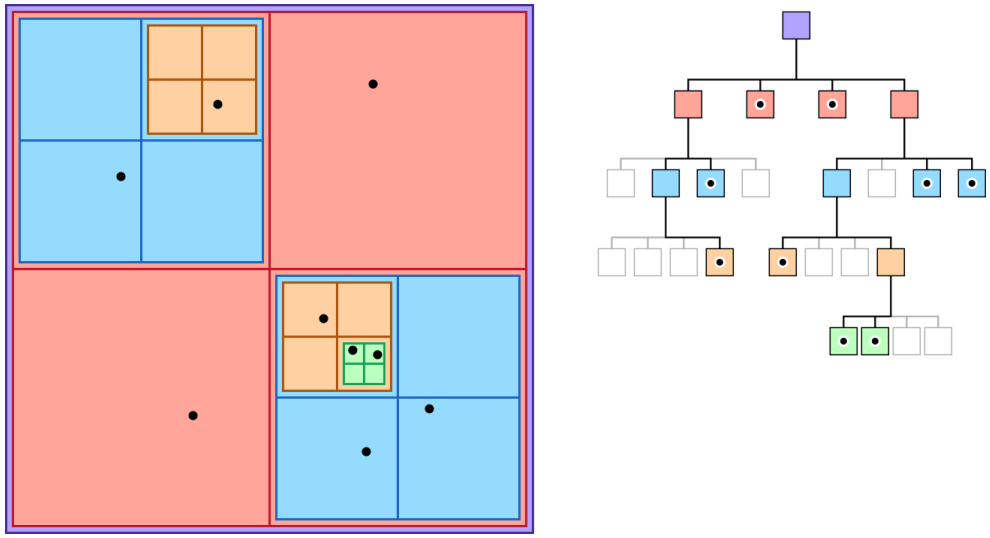

Figure 5: Points on the plane and a corresponding quadtree, courtesy of [App20].

contains singleton subsets of X. In other words, $P_0 = \{\{x\} : x \in X\}$. This is a *hierarchical partition* of (X, d). We say $\mathcal{P}_i$ is the *i-th level* of the hierarchical partition.

A split-tree of (X, d) follows naturally from a hierarchical partition.

Definition 3.2 (Split-Tree): Given a hierarchical partition $\mathcal{P} = \mathcal{P}_\ell, \mathcal{P}_{\ell-1}, \ldots, \mathcal{P}_0$, the corresponding *split-tree* is as follows:
The root is node is X. Notice how the root is a cluster on level ℓ.
Recursively, if the current node represents a cluster C on level i, then its children are the clusters on level $i-1$ which partition C.

We now construct a split-tree with "nice" properties. The embedding of (X, d) follows quickly from this split-tree.

3.1.1 Nested Covers

First, we require a small subroutine. Without loss of generality, by scaling if necessary, we may assume that the minimum distance between points is 1. Let Δ be the aspect ratio. Note that Δ is really the diameter of (X, d).

Suppose `Greedy_Net(Y, d, delta)` is the subroutine that greedily constructs a δ-net of some $Y \subseteq (X, d)$.

```
def Nested_Covers(X, d):
  Y[0] := X

  for i = 1, 2, ..., log_2(Delta):
    Y[i] := Greedy_Net(Y[i-1], d, 2**(i-2))

  return Y
```

`Nested_Covers` returns a nested sequence $X = Y_0 \supseteq Y_1 \supseteq \cdots \supseteq Y_{\lceil \log_2 \Delta \rceil}$ such that Y_i is a 2^{i-2}-net of Y_{i-1}. Remark that Y_1 is a 2^{-1}-net of X. Moreover, Y_2 is a 2^0-net of Y_1. By the triangle inequality, any point of X is of distance at most 2^1 from Y_2, thus Y_2 is a 2^1-cover of X. Inductively, Y_i is a 2^{i-1}-cover of X. Following a similar argument, we can show that the minimum distance of Y_i is strictly greater than 2^{i-3}.

3.1.2 Split-Tree Construction

The construction of Talwar's randomized split-tree requires two sources of randomness. Both of which are crucial to the analysis, as we will see.

Choose ρ uniform randomly from $\left[\frac{1}{2}, 1\right)$ and π uniform randomly from all permutations of X. We will construct a hierarchical partition of X and omit for the sake of simplicity the pseudocode for actually constructing the split-tree. It is not too hard to build the split-tree while building the hierarchical partition.

```
def Hierarchical_Partition(X, d):
  Y := Nested_Covers(X, d)
  P[log_2(Delta) + 1] := {X}

  for i = log_2(Delta), ..., 0:
    Y[i].sort(pi) # sort with respect to random permutation
    r_i := (2**i) * rho
    for x in X : assigned[x] = False

    for C in P[i+1]:
      for y in Y[i]:
        new_cluster := {x in C : d(x, y) <= r_i and not assigned[x]}
        for x in new_cluster: assigned[x] = True
        if new_cluster is not None: P[i].add(new_cluster)

  return P
```

In words, for each cluster $C \in \mathcal{P}_{i+1}$, each $y \in Y_i$ comes in order of π and cuts out a new cluster consisting of unassigned points [1]. See Figure 6 and Figure 7. Note that it is entirely possible that y does not belong to the new cluster if it does not originally belong to C but covered some points of C. It is also plausible that each $y \in Y_i$ cuts out multiple clusters.

3.1.3 Analysis

We now show the following of Talwar's split-tree.

[1] Talwar's original pseudocode omitted the sorting step, which makes the construction a bit more confusing.

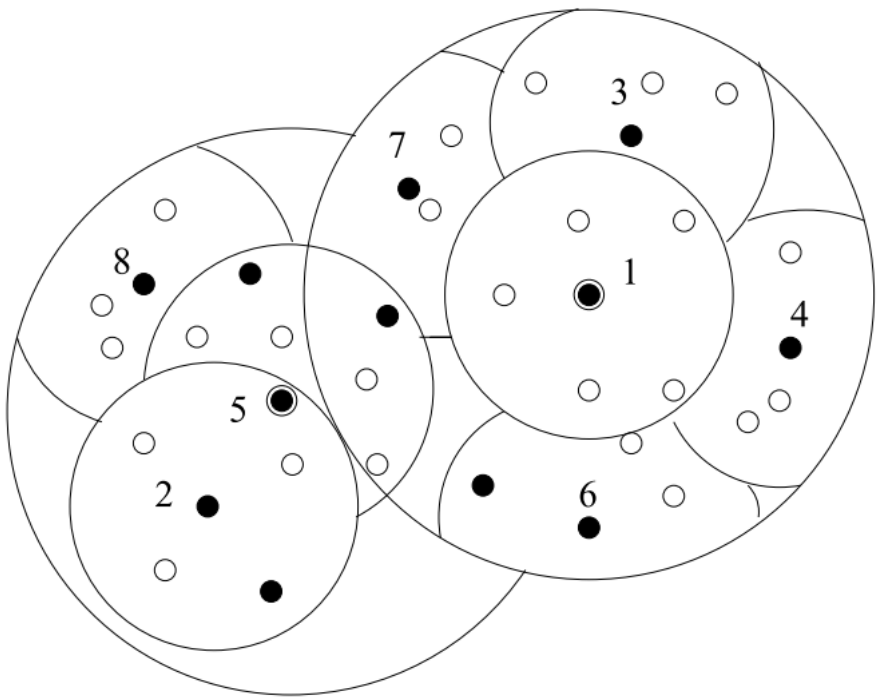

Figure 6: The hierarchical partition of a metric space labelled according to a random permutation π, courtesy of [Tal04].

Proposition 3.2:
`Hierarchical_Partition` returns a hierarchical partition with $2 + \lceil \log_2 \Delta \rceil$ levels.

Proof (Proposition 3.2): It is clear that the top level is the trivial partition and $\mathcal{P}_i$ is a union of partitions of $\mathcal{P}_{i+1}$. We need only show that $\mathcal{P}_0$ consists of singletons. The number of levels follows.
Indeed, observe that by construction, $r_0 < 1$, which is strictly less than the minimum distance between points of X. In particular, for all $y \in Y_0 = X$,

$$d(x, y) \leq r_0 \iff x = y.$$

Thus $\mathcal{P}_0$ indeed consists of singletons and `Hierarchical_Partition` returns a hierarchical partition of (X, d).$\square$

Proposition 3.3:
Each level i cluster has diameter at most 2^{i+1}.

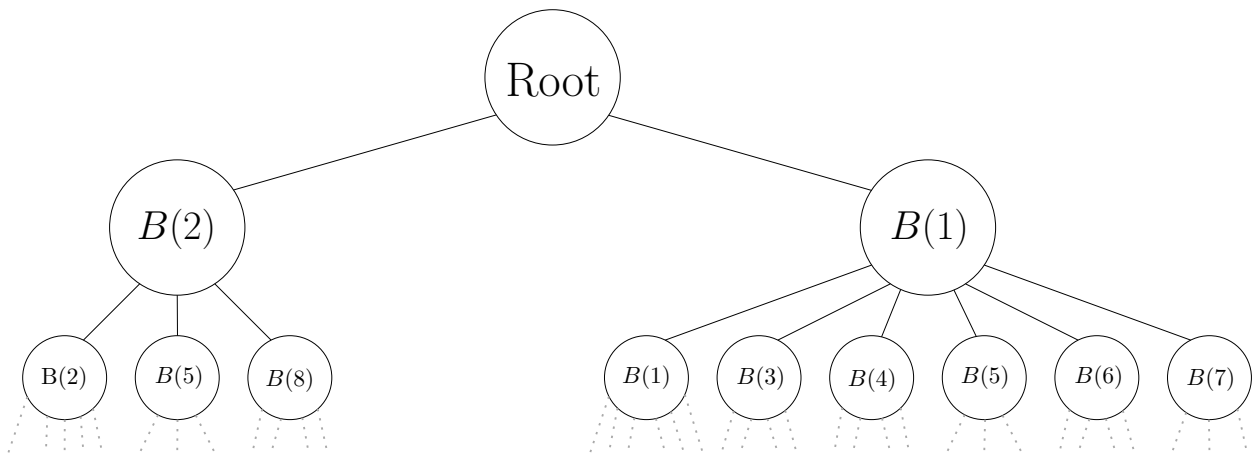

Figure 7: The accompanying split-tree to Figure 6 labelled with the balls using which clusters were "cut".

Proof (Proposition 3.3): This holds for the top level $(1 + \lceil \log_2 \Delta \rceil)$ since the diameter of X is

$$2^{\log_2 \Delta} \leq 2^{\lceil \log_2 \Delta \rceil + 2}.$$

For every other level, each cluster C was cut out by some $y \in Y_i$ so $C \subseteq B_{r_i}(y)$. Clearly, the diameter of C is at most $2r_i \leq 2^{i+1}$. □

Proposition 3.4:
Each level i cluster is the union of at most $2^{O(k)}$ level $i-1$ clusters.

Proof (Proposition 3.4): Consider a level $i+1$ cluster $C \subseteq B_{r_{i+1}}(z)$ for some $z \in Y_{i+1} \subseteq Y_i$. Any $y \in Y_i$ such that $B_{r_i}(y) \cap C \neq \varnothing$ must satisfy

$$d(y, z) \leq r_i + r_{i+1} \leq 3 \cdot 2^i.$$

Moreover, we showed that the minimum distance between points of Y_i is strictly greater than

$$2^{i-3}.$$

By Proposition 2.4,

$$|\{y \in Y_i : B_{r_i}(y) \cap C \neq \varnothing\}| \leq 2^{k \cdot O(\log_2 3 \cdot 2^3)} = 2^{O(k)}$$

as desired. □

It is not an exaggeration to say that Theorem 3.5 is the most crucial result of Talwar's paper [2]. Our analysis of the distortion and treewidth of Talwar's embedding depends heavily on it.

[2]Talwar provided a sketch proof of Lemma 3.6. He omitted the argument of infinite series, which confused me to no end.

Theorem 3.5:
For any pair of points $u, v \in X$, the probability that u, v fall into different level i_0 clusters is at most

$$O(k)\frac{d(u,v)}{2^{i_0}}.$$

Lemma 3.6:
For any two points $u, v \in X$ which are in the same cluster on level $i+1$, the probability of u, v falling in different clusters on level i is at most

$$O(k)\frac{d(u,v)}{2^{i}}.$$

Proof (Lemma 3.6): Sort the points $y \in Y_i$ in increasing order of

$$\min\{d(u,y), d(v,y)\}.$$

Say $Y_i = \{y^{(1)}, y^{(2)}, \ldots\}$. Notice that this is a permutation of Y_i.
We say some $y \in Y_i$ *cuts* the pair (u, v) if exactly one of u, v falls in

$$B_{r_i}(y)$$

and

$$B(z, r_i) \cap \{u, v\} = \varnothing$$

for all $z \in Y_i$ such that $\pi(z) < \pi(y)$.
Clearly, if u, v falls into different level i clusters while being in the same cluster on level $i+1$, such a cutter exists. We will bound the probability of the existence of such a cutter.
Suppose that $y^{(j)}$ cuts u, v.
Without loss of generality, by relabeling if necessary, $u \in B_{r_i}(y^{(j)})$. Thus we have

$$r_i \in \left[d(y^{(j)}, u), d(y^{(j)}, v)\right).$$

This happens with probability at most

$$\begin{aligned}
&\int_{d(y^{(j)},u)}^{d(y^{(j)},v)} \frac{1}{2^i - 2^{i-1}}dt && r_i \in [2^{i-1}, 2^i) \\
&= \frac{d(y^{(j)}, v) - d(y^{(j)}, u)}{2^{i-1}} \\
&\leq \frac{d(u,v)}{2^{i-1}}. && \text{reverse triangle inequality}
\end{aligned}$$

Here the integral can be taken to either be the Riemann or Lebesgue integral.
Conditioned on the above, we must also have

$$\forall k < j, \pi(y^{(j)}) < \pi(y^{(k)}).$$

Viewing π as a permutation of $y^{(1)}, y^{(2)}, \ldots, y^{(j)}$, we see that this happens with probability

$$\frac{1}{j}.$$

All in all

$$P(y^{(j)} \text{ cuts } u, v) \leq \frac{d(u,v)}{2^{i-1}} \cdot \frac{1}{j}.$$

Remark that the statement trivially holds when $d(u,v) \geq 2^i$, thus we may assume without loss of generality that $d(u,v) \leq 2^i$. Observe that for some $y^{(j)}$ to cut u, v, then it must be within distance 2^i of either u or v. Let

$$Y_i' := \{y \in Y_i : \min\{d(y,u), d(y,v)\} \leq 2^i\}.$$

The maximum distance between points of Y_i' is at most $3 \cdot 2^i$. On the other hand, we showed the minimum distance between points of Y_i is 2^{i-3}. It follows by the ever-useful Proposition 2.4 that

$$|Y_i'| \leq 2^{k \cdot O(\log_2 3 \cdot 2^3)} = 2^{O(k)}.$$

Hence the overall probability of u, v being cut is at most

$$\begin{aligned}
\sum_{y^{(j)} \in Y_i'} P(y^{(j)} \text{ cuts } u, v) &\leq \sum_{j=1}^{2^{O(k)}} \frac{d(u,v)}{2^{i-1}} \cdot \frac{1}{j} \\
&= \frac{d(u,v)}{2^{i-1}} \log 2^{O(k)} && \text{harmonic series} \\
&\leq O(k) \frac{d(u,v)}{2^i}
\end{aligned}$$

as required. □

The proof of Theorem 3.5 follows immediately from Lemma 3.6.

Proof (Theorem 3.5): The probability that u, v fall in separate of clusters on level i_0 is the probability that some $y \in Y_i, i \geq i_0$ cuts u, v. By a union bound, this is at most

$$\begin{aligned}
\sum_{i=i_0}^{\lceil \log_2 \Delta \rceil + 1} O(k) \frac{d(u,v)}{2^i} &\leq O(k) d(u,v) \sum_{i=i_0}^{\infty} 2^{-i} \\
&= O(k) d(u,v) \cdot \frac{2^{-i_0}}{1 - \frac{1}{2}} \\
&= O(k) \frac{d(u,v)}{2^{i_0}}.
\end{aligned}$$

□

3.2 The Algorithm

The idea of the embedding follows naturally from the split-tree [3]. We will replace each cluster with a suitably chosen net called *portals* to form a tree decomposition of some graph $H = (X, E)$.

More rigorously, we construct a tree decomposition of (T, I) of our embedding. The underlying host tree T of our embedding is precisely the split-tree. The embedding $H = (X, E)$ is determined exactly by the tree decomposition. Indeed, $uv \in E$ if and only if uv is covered by some bag. In other words, H **has a clique**

[3] We note that Talwar did not give the explicit embedding. Thus the embedding we give may have deviated slightly from the author's original designs but still attains the same bounds.

on every bag of the tree decomposition. We give such an edge uv the weight

$$w(uv) := d_X(u, v).$$

Let $\beta > 0$ be a constant to be chosen later. `Split_Tree(P)` is the subroutine which yields the split-tree corresponding to a hierarchical partition $\mathcal{P}$. `Greedy_Net(C, Z, beta, i)` is the subroutine which takes a cluster C on level i, some $Z \subseteq C$ with minimum distance $\beta 2^i$, and greedily builds a $\beta 2^i$-net of C beginning with the well-separated points Z.

```
def Talwar_Embedding(X, d):
  P := Hierarchical_Partition(X, d)
  T := Split_Tree(P)

  def Recurse(C, Z, i):
    new_Z := Greedy_Net(C, Z, beta, i)
    C.bag = (Z_0)
    for child in C:
      Recurse(child, new_Z and child, i-1) # new_Z intersect child

  Recurse(T.root, {}, 1 + log_2(Delta)) # top level cluster, empty set
  return T
```

In words, we recursively build (T, I) top-down. Assuming we have chosen the appropriate portals for the parent cluster, greedily build a net of the current cluster starting with the inherited portals. This decomposition clearly covers all vertices since the leaves of the split-tree are singletons. Moreover, we dictate which edges exist in the embedding thus there is no need to worry about covering edges. Finally, the connectivity condition follows by our choice of portals. If a point $v \in X$ was first chosen to be a portal on level i, then it will be a portal for some cluster on levels $i-1, i-2, \ldots, 0$ and so the connectivity condition is always satisfied. In fact, the subgraph induced by nodes containing v is a subpath of T.

3.3 Analysis

Let us analyze the treewidth and expected distortion of our embedding. The key is to choose an appropriate parameter $\beta > 0$ for our algorithm.

3.3.1 Expected Distortion

We can think of our embedding in terms of the complete graph representation of (X, d). Essentially, we are sparsifying the edges based on whether their endpoints are both portals. It follows that distances in the embedding can only grow. Since the metric of the embedding H is based on shortest paths, for an arbitrary pair $u, v \in X$ we will describe some path in H which is at most $O(\epsilon) d_X(u, v)$ longer than $d_X(u, v)$ in expectation. This yields an upper bound on the distortion.

Lemma 3.7:
For all $u, v \in V$, there is a uv-path in H with expected length at most

$$(1 + O(\beta k \log_2 \Delta)) \cdot d_X(u, v).$$

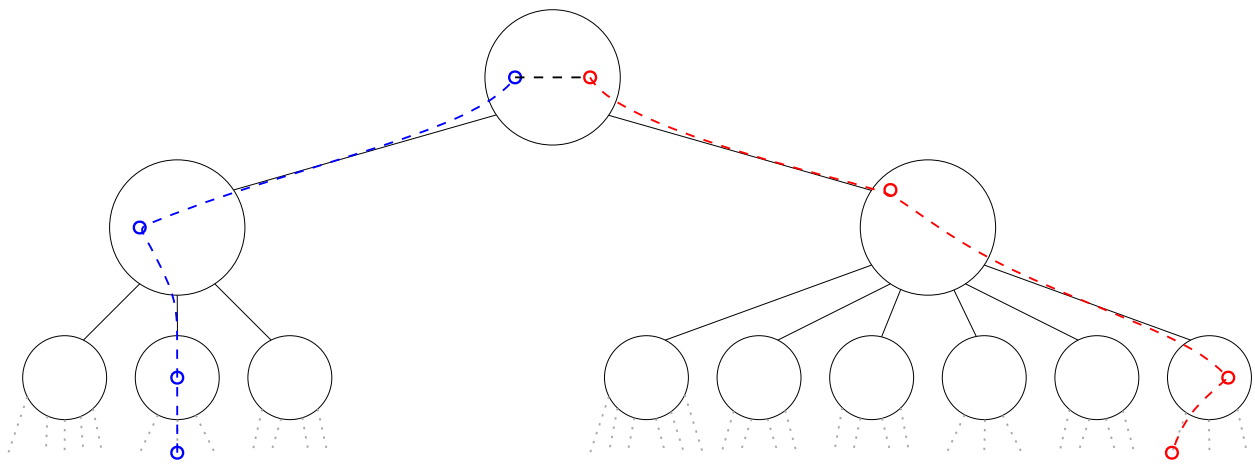

Figure 8: A uv-path of portals in the embedding obtained from the split-tree in Figure 7.

Proof (Lemma 3.7): The idea is to start from level 0 of the split-tree and travel "upwards" along portals until the two paths reach the same cluster. See Figure 8.
On each level $i \geq 0$, there are portals u_i, v_i which are at most $\beta 2^i$ from u, v, where $u_0 := u, v_0 := v$. Observe that

$$d_H(u_i, u_{i+1}) \leq d_X(u_i, u) + d_X(u, u_{i+1}) \leq 2\beta 2^{i+1}$$

and likewise for $d_H(v_i, v_{i+1})$. Thus given that u_i, v_i reside in different clusters on level i, traveling from level i to $i+1$ incurs a cost of $8\beta 2^i$.
Let $\ell \leq 1 + \lceil \log_2 \Delta \rceil$ be the level for which u_ℓ, v_ℓ reside in the same cluster. There is an edge between u_ℓ, v_ℓ by construction so that

$$d_H(u_\ell, v_\ell) \leq d_X(u_\ell, u) + d_X(u, v) + d_X(v, v_\ell) \leq 2\beta 2^\ell + d_X(u, v).$$

At each level $0 < i \leq 1 + \log_2 \Delta$, the probability that u_i, v_i reside in different clusters is at most $O(k)\frac{d_X(u,v)}{2^i}$ by Theorem 3.5. Thus by the linearity of expectation, the expected length of the described path is at most

$$\begin{aligned} d_X(u, v) + \sum_{i=1}^{1+\lceil \log_2 \Delta \rceil} O(k)\frac{d_X(u, v)}{2^i} \cdot 10\beta 2^i &= d_X(u, v) + (1 + \lceil \log_2 \Delta \rceil) O(k) \cdot 10\beta d_X(u, v) \\ &\leq (1 + O(k\beta \log_2 \Delta)) \cdot d_X(u, v). \end{aligned}$$

□

Following Lemma 3.7, the choice of β becomes clear. Taking

$$\boxed{\beta := \frac{\epsilon}{O(k \log_2 \Delta)}}$$

yields the desired bound on expected distortion.

3.3.2 Treewidth

Having chosen β, let us now examine the width of the computed tree decomposition. Recall from Proposition 2.4 that the aspect ratio of a subset of X bounds its cardinality.

Fix $0 \leq i \leq 1 + \lceil \log_2 \Delta \rceil$. By definition, the minimum distance of a $\beta 2^i$-net of a cluster on level i is at least $\beta 2^i$. Moreover, the maximum distance is at most the diameter of the cluster, which is at most 2^{i+1} from Proposition 3.3. Thus the aspect ratio of our net is at most

$$\frac{2^{i+1}}{\beta 2^i} = \frac{2}{\beta}.$$

Remark this holds independently of our choice of i.

It follows by Proposition 2.4 that the width of the computed embedding is at most

$$2^{k \cdot O(\log_2 \frac{2}{\beta})} = 2^k \cdot \left(\frac{O(k \log \Delta)}{\epsilon} \right) \in O\left(2^k \cdot \frac{k \log \Delta}{\epsilon} \right)$$

according to our choice of β.

4 Approximation

Talwar described quasi-polynomial approximation schemes (QPTAS) for several problems including Metric TSP, Steiner Tree, and Facility Location directly using the split-tree. We will give several QPTAS for metric TSP leveraging the tree decomposition and defer the interested reader to [Tal04] and [Fel+18]. The order in which we give these schemes is not chronological but in order of running time. Our goal is to first provide a broader overview of techniques to solve problems using the tree decomposition and then give specialized algorithms with faster run-times specifically for metric TSP.

Definition 4.1 (Quasi-Polynomial): A *quasi-polynomial* is a function $\mathbb{R} \to \mathbb{R}$ of the form

$$2^{O(\log^c n)}$$

for some fixed $c > 0$.

In general, quasi-polynomial time algorithms are faster than exponential time algorithms and can be slower can polynomial time algorithms. For the special case of $c = 1$, we have a polynomial. On the other hand, for $c < 1$, the function is sublinear. When we refer to quasi-polynomial time algorithms, it is usually implicit that $c > 1$.

Problem 1 (Metric TSP): For a finite metric space (X, d), a *tour* is a sequence of points

$$x_0, x_1, \ldots, x_\ell = x_0$$

such that every point of X appears at least once. Notice we allow for repeats.
The length of this tour is

$$\sum_{j=0}^{\ell-1} d(x_j, x_{j+1})$$

Compute a minimal length tour.

The high-level idea is simple. We compute an exact solution in the embedding H and since expected distortion is arbitrarily small, we can make our exact solution in H arbitrarily close to an optimal solution of (X, d). The run-time to obtain an exact solution in H depends on the width of our decomposition.

However, the treewidth bound from Theorem 3.1 is small only if the aspect ratio is small. In particular, we can make the aspect ratio exponential with respect to the input metric space. To work around this, we compute a coarse subset with small aspect ratio and show it has doubling dimension at most $2k$. This is the actual set we embed. See [Fel+18] for similar techniques.

4.1 Bounding the Aspect Ratio

To bound the aspect ratio, We take a detour to recall a basic $O(1)$-approximation algorithm for metric TSP.

Proposition 4.1:
There is a polynomial time computable $O(1)$-approximation of metric TSP.

See [WS11] for details. The idea is that the shortest tour in (X, d) is at least as long as a minimum spanning tree T. Our approximate tour essentially traverses T twice to give a 2-approximation. See figure 9. The approximation guarantee can actually be improved to $\frac{3}{2}$ by solving a minimum weight perfect matching problem. See [Chr76] for more details. However, we only need a constant approximation.

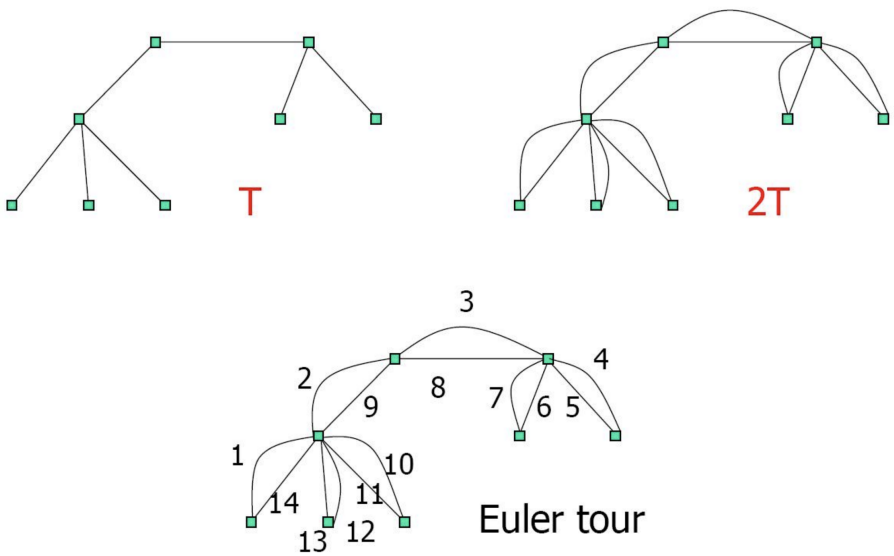

Figure 9: The MST based 2-approximation of metric TSP, courtesy of [CI11].

Let OPT be the weight of an optimal tour in (X, d). We can obtain an approximation L of OPT such that $L \in \Theta(\text{OPT})$ by proposition 4.1. Greedily compute a

$$\frac{L\epsilon}{n}\text{-net}$$

of (X, d) and call it Y.

By construction, the minimum distance between any two points of Y is at least $\frac{L\epsilon}{n}$. To obtain an upper bound on the maximum distance, observe that a tour contains a path of length at most OPT between any two vertices. It follows that the maximum distance between any two points of Y is at most L. The aspect ratio of N is then bounded above by

$$L \cdot \frac{n}{L\epsilon} = \frac{n}{\epsilon}.$$

Lemma 4.2:
(Y, d) has doubling dimension at most $2k$.

Proof (Lemma 4.2): Fix $y \in Y, r > 0$ and put $B_{X,r}(y)$ as the ball of radius r in (X, d).
Apply the definition of doubling dimension twice. There is some $Z \subseteq B_{X,r}(y)$ such that

$$B_{X,r}(y) = \bigcup_{z \in Z} B_{X,r/4}(z)$$

with cardinality $|Z| \leq 2^k \cdot 2^k = 2^{2k}$.
For every $z \in Z$ such that $B_{X,r/4}(z) \cap Y \neq \varnothing$, pick some $y_z \in B_{X,r/4}(z) \cap Y$ and observe that

$$B_{X,r/4}(z) \subseteq B_{X,r/2}(y_z).$$

It follows that

$$B_{X,r}(y) \subseteq \bigcup_{z \in Z} B_{X,r/2}(y_z).$$

The containment is preserved when we consider balls of the same radius within (Y, d), concluding the proof.□

By Theorem 3.1, our work above, and Lemma 4.2, Talwar's embedding computes a tree decomposition with width at most

$$\boxed{\log^{O(1)} n}$$

assuming we fix ϵ and the doubling dimension. This induces an embedding H_Y of (Y, d) with expected distortion at most $1 + O(\epsilon)$.

4.2 Obtaining a Solution in the Net

Now we have an embedding of our net and a tree decomposition with small width, we are ready to compute exact solutions in the embedding.

4.2.1 Monadic Second-Order Logic

Many problems can be solved with dynamic programming using the tree decomposition of a graph. The book by Courcelle and Engelfriet [CE12] presents a unified method. Intuitively, if a property of a graph can be described with a logic formula that is "not too big" we can evaluate this formula bottom-up using a tree decomposition.

The language of valid formulas is called Extended Monadic Second-Order Logic (EMSOL2) and is beyond the scope of this survey. See also [ALS91]. For example, the statement "there is a tour of length L" can be described as

$$\begin{aligned}
&\exists C \subseteq E \\
&\neg \text{Disconnected}(H[C]) \\
&\wedge \text{MaxDegree}(H[C]) = 2 \\
&\wedge \text{MinDegree}(H[C]) = 2 \\
&\wedge \sum_{e \in C} w(e) = L.
\end{aligned}$$

Each of these "functions" can be explicitly expressed with basic logic symbols.

If the width of the tree decomposition is $\mathcal{K}$. Then there is a computable function f dependent only on the treewidth such that we can solve the decision variant of metric TSP in

$$O(f(\mathcal{K})n)$$

time. The optimization version can then be solved using binary search.

4.2.2 Dynamic Programming

The source of inspiration for EMSOL2 was the myriad of dynamic programming solutions for NP-hard problems using the tree decomposition. The details of such techniques are once again beyond the scope of this survey. See [Bie20] for exact details.

Essentially, we are solving smaller instances of the problem at subtrees of the host tree and combining them in a meaningful manner. The resulting run-time can be shown to be

$$O\left(\mathcal{K}^{\mathcal{K}} n\right).$$

Plugging in the treewidth bound we computed earlier actually yields quasi-polynomial running times. However, there are even faster recent improvements.

4.2.3 Probability & Linear Algebra

There is a recently developed randomized algorithm which attains a running time of

$$4^{\mathcal{K}} n^{O(1)}$$

using the so-called "Cut&Count" technique [Cyg+11]. This has been improved to a deterministic algorithm at the cost of a slower run time of

$$\widetilde{O}(15^{\mathcal{K}} \mathcal{K}^{O(1)} n^2)$$

using a linear algebraic approach [Bod+13]. The $\widetilde{O}$ notation hides a polylogarithmic factor. In addition to being of innovative interest, these algorithms immediately show us that we can solve for the optimal tour in quasi-polynomial time.

4.3 Obtaining an Approximate Solution

Given a solution S in H_Y, we wish to obtain an approximate solution in (X, d). We can simply reuse the same tour in (X, d). The only caveat is that there might be unvisited points of $X \setminus Y$.

For each $u \in Y$, if there is an unvisited point $v \in X \setminus Y$ which falls in

$$B_{L\epsilon/n}(u),$$

then travel to v, return to u, and repeat if necessary. See figure 10. Recall that that Y is a $\frac{L\epsilon}{n}$-net of (X, d). With these detours, our tour clearly visit all points in X.

Call the tour W. Obtaining W from S incurs a cost of at most

$$2n\frac{L\epsilon}{n} = 2L\epsilon.$$

The optimal tour S in H_Y has cost at most OPT since the optimal tour in (X, d) contains Y as a subset. Moreover, the cost function of a tour is linear in the edge costs, meaning that we may apply the linearity of expectation. Specifically, the weight of S within (X, d) is at most

$$(1 + O(\epsilon))\left(\sum_{uv \in E(S)} d_X(u, v)\right) \leq (1 + O(\epsilon))\,\mathrm{OPT}$$

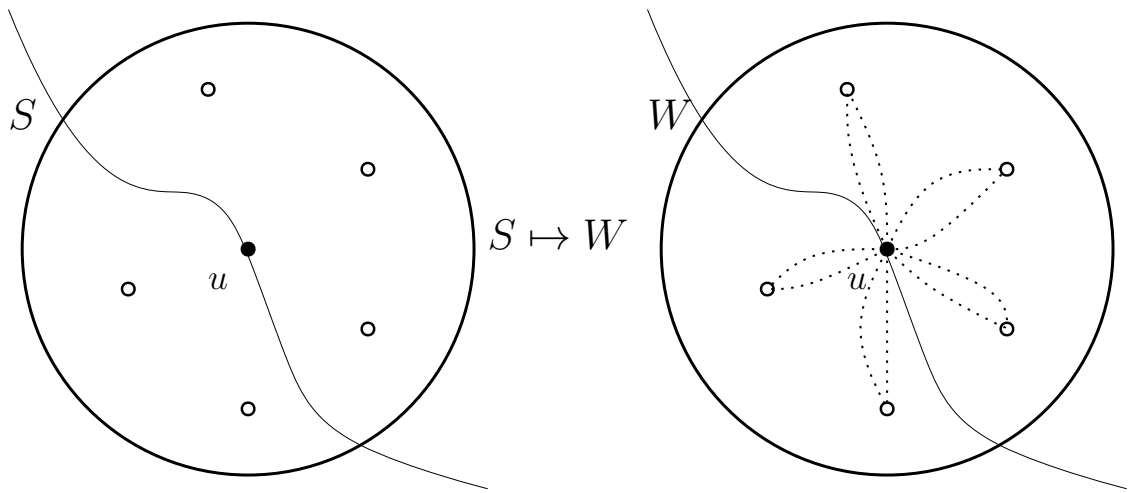

Figure 10: Obtaining W from S at a single point $u \in Y$.

in expectation. Factoring in the additional penalty incurred from modifying S to W and the fact that $L \in \Theta(\mathrm{OPT})$, we see the expected weight of W is at most

$$(1 + O(\epsilon))\,\mathrm{OPT} + O(\epsilon)\,\mathrm{OPT}\,.$$

This yields the approximation guarantee.

4.4 Repeated Trials

The astute reader will recognize that we have only shown that the expected weight of W attains the desired approximation guarantee of $1 + \epsilon$. We need only apply a common trick to attain this bound with high probability. Indeed, let $\mathcal{C}$ be the random variable denoting the cost of W. Recall that Markov's Inequality states that

$$\begin{aligned} P(\mathcal{C} \geq (1 + 2\epsilon)\,\mathrm{OPT}) &\leq \frac{E[\mathcal{C}]}{(1 + 2\epsilon)\,\mathrm{OPT}} \\ &\leq \frac{1 + \epsilon}{1 + 2\epsilon} \\ &= 1 - \frac{\epsilon}{1 + 2\epsilon} \\ &\leq 1 - \epsilon. \end{aligned}$$

By convention, "high probability" means that the probability of failure is at most $\frac{1}{4}$, thus that is our desired goal. Let t be the variable representing the number of independent trials we run our algorithm. We want

$$(1 - \epsilon)^t \leq \frac{1}{4}.$$

We will employ the standard inequality $1 - x \leq e^{-x}$. Our bound then becomes

$$\begin{aligned}(1-\epsilon)^t &\leq e^{-t\epsilon} \leq \frac{1}{4} \\ -t\epsilon &\leq -\ln 4 \\ t &\geq \frac{\ln 4}{\epsilon} \\ &\in O\left(\frac{1}{\epsilon}\right).\end{aligned}$$

Thus with $O\left(\frac{1}{\epsilon}\right)$ repetitions, our algorithm attains the approximation guarantee with high probability.

Theorem 4.3:
Let (X, d) be a metric space of doubling dimension k.
For any $\epsilon > 0$, there is a randomized algorithm that computes a $1 + \epsilon$ approximate tour of X with high probability in quasi-polynomial time.

The running time of our algorithm is dominated by the running time to solving for an exact solution in the embedding of a net. Indeed, a net can be computed in linear time, Theorem 3.1 computes an embedding in polynomial time, while extending an exact solution in the embedding to the approximation solution can be done in polynomial time.

4.5 Other Problems

We conclude this section by emphasizing an earlier remark. By adopting similar steps taken in the QPTAS we provided, it is possible to obtain QPTAS for other "distance" related problems such as the Steiner Tree and Facility Location problems. This reveals the importance of interpreting Talwar's result in terms of treewidth as we can use directly use existing algorithms in our approximation scheme instead of re-inventing the wheel.

There may be slight hiccups in the translation between schemes but the key ideas have all been described: Bound the aspect ratio, compute an optimal solution in the embedding, and fiddle with it to obtain an approximate solution in (X, d).

5 Miscellaneous Results

In this section, we illustrate some additional results and our unsuccessful attempt to derive a connection between the metrics of constant doubling dimension and the Cantor set.

5.1 A PTAS for TSP

In a recent paper, Bartal et al. extended Talwar's work to a PTAS for TSP in metric spaces of bounded doubling dimension [BGK16]. Specifically, the authors showed that a $(1 + \epsilon)$-approximation to the optimal tour in a metric space of doubling dimension k on n points can be computed by a randomized algorithm in

$$n^{2^{O(k)}} 2^{(2^k/\epsilon)^{O(k)}} \sqrt{\log n}$$

time.

The baseline for this work is a carefully-chosen variant of Talwar's original algorithm using the split-tree. Gupta et al. approximated the cost incurred by an optimal tour within individual clusters of the split-tree and paid extra attention to expensive clusters. Recall that Arora's original approximation scheme [Aro98]

is restricted to Euclidean spaces of bounded dimension. This result completely frees the restriction from Euclidean spaces.

5.2 Connections to the Cantor Set

The following is a small exploration of the connection between the Cantor set and metric spaces of constant doubling dimension. We require some non-trivial analysis facts which can be found in Rudin's classic book [Rud+64]. Many of the definitions we sketch are not completely rigorous and are only meant to convey an idea of the notions.

5.2.1 The Cantor Set

Mathematically, the Cantor set is an extremely popular counterexample to many statements. It has jokingly been said that "almost all" [4] counterexamples in analysis is or can be derived from the Cantor set.

We define the Cantor set by recursively removing "middle thirds" from $[0, 1]$. Specifically, let $C_0 := \{[0, 1]\}$. If $C_i = \{[a, b], \ldots\}$ is a collection of disjoint closed intervals, we define

$$C_{i+1} := \left\{ \left[a, a + \frac{b-a}{3}\right], \left[b - \frac{b-a}{3}, b\right], \ldots \right\}.$$

The Cantor set is then the intersection of all such sets

$$C := \bigcap_{i \geq 0} \bigcup_{I \in C_i} I.$$

We give it the same metric inherited from $\mathbb{R}$. Namely,

$$d(x, y) = |x - y|.$$

Notice how we can "parse" C into an infinite binary tree. The i-th level nodes represent C_i and its children are the two smaller intervals resulting from removing the middle thirds. This is already very similar to Talwar's split-tree. See Figure 11.

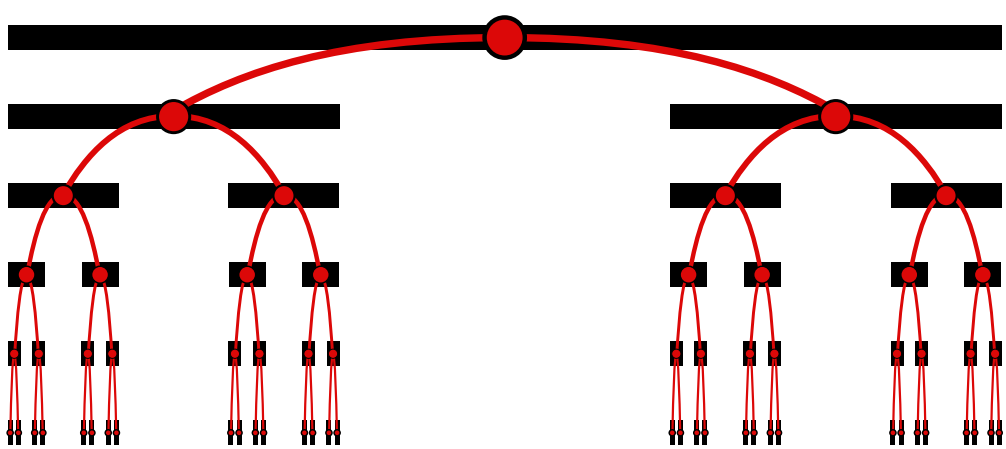

Figure 11: The Cantor set and corresponding "parse-tree", courtesy of [Der16].

[4] A measure theory pun.

5.2.2 Compactness

The notion of *compactness* is extremely powerful. Roughly speaking, it generalizes closedness and boundedness in $\mathbb{R}^k$. We avoid defining open and closed sets here. The Borel-Lebesgue theorem provides many equivalent characterizations of compactness in a metric space.

For instance, a corollary of the Borel-Lebesgue theorem is that any finite metric space is compact. It can also be shown that the Cantor set is compact.

Perhaps the most popular use of compact sets is the Extreme Value Theorem.

Theorem 5.1 (Extreme Value):
A real-valued function $f : C \to \mathbb{R}$ on a compact set attains its minima and maxima.

This result generalizes the commonly taught case where $C = [a, b]$ is a closed interval in $\mathbb{R}$.

5.2.3 Total Disconnectedness

Roughly speaking, a subset $Y \subseteq X$ of the metric space (X, d) is *disconnected* if there are two disjoint open sets of X containing Y. The *connected component* of $x \in X$ is the largest connected subset of X containing x. A metric space is *totally disconnected* if the connected component of every point is just itself. The canonical example of a totally disconnected set is the Cantor set. See also Figure 12.

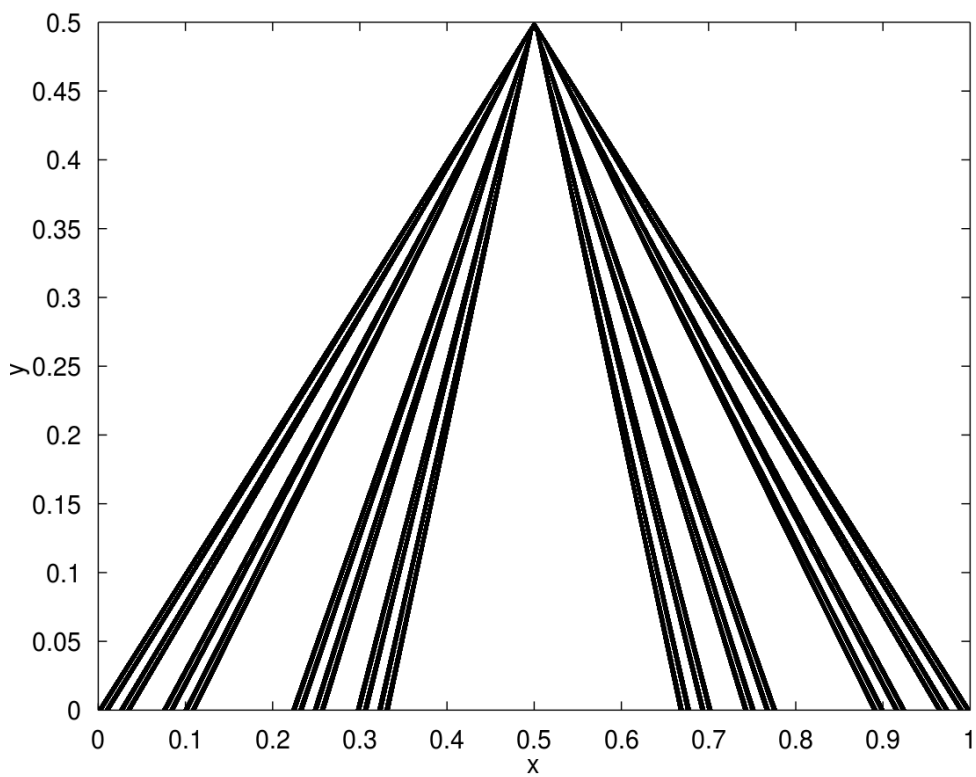

Figure 12: The Knaster-Kuratowski Fan is a connected metric space which becomes totally disconnected from the removal of a single point, courtesy of [Gkn14].

It is important to make the distinction between similar definitions of graph theory and analysis. Indeed, any finite metric space is totally disconnected since there is a minimum distance r between any two points. Thus a ball of radius $r/4$ around every point is a "certificate" that our metric space is totally disconnected. Hence the shortest path metric space of a finite connected graph is totally disconnected.

5.2.4 A Dead-End

There is one additional definition we need. A metric space is *perfect* roughly speaking if for all $x \in X$

$$\inf_{y \in X \setminus \{x\}} d(x, y) = 0.$$

Thus we can get arbitrarily close any point of X with other points.

The following theorem is a surprising fact.

Theorem 5.2:
If (X, d) is compact, perfect, and totally disconnected, then there is a continuous bijection

$$\phi : X \to C$$

such that ϕ^{-1} is also continuous.
Here C denotes the Cantor set.

The proof essentially shows that we can parse (X, d) similarly to the Cantor set and "matches" their parsings. It suggests we may be able to do something similar to Talwar's split-tree.

Unfortunately, this theorem does not apply to our finite metric spaces since any finite metric space is never perfect for the same reason it is totally disconnected. Fortunately, there is a slightly weaker result [Kar] which relaxes the condition of perfectness.

Theorem 5.3:
If (X, d) is compact and totally disconnected, then there is a continuous bijection

$$\phi : X \to B \subseteq C$$

such that ϕ^{-1} is also continuous.
Here B is a compact subset of C.

Notice how Theorem 5.3 applies to any finite metric space.

The reason our exploration is at a dead-end is that we have yet to use any properties of doubling dimension. Everything has come for free from the fact that our metric spaces are finite. Perhaps if we allow for infinite metric spaces with constant doubling dimension such as $\mathbb{R}^k$, we would be able to make more progress. However, a key requirement of Talwar's split-tree is that there is a "top"-level from which we can begin our parsing. For infinite metric spaces of constant doubling dimension, we would need our metric spaces to be bounded. Due to time limitations, we conclude our exploration here.

We end this section by directing the interested reader to [PW11] and [WWW15] for somewhat related discussions.

6 Conclusion

We have shown through Talwar's work that some important algorithmic and structural properties of Euclidean spaces are shared by metric spaces of bounded doubling dimension. Most notably, we have shown

the existence of a probabilistic QPTAS for metric TSP in these particular metric spaces. Recent extensions have improved the running time to polynomial.

On two brief detours, we explore general methods to solve problems using the tree decomposition and potential connections between the doubling dimension and the Cantor set.

We conclude this survey by challenging the reader to extend the techniques we employed to solve metric TSP and solve other problems.

7 References

[ALS91] Stefan Arnborg, Jens Lagergren, and Detlef Seese. "Easy problems for tree-decomposable graphs". In: *Journal of Algorithms* 12.2 (1991), pp. 308–340. DOI: 10.1016/0196-6774(91)90006-k.

[App20] Apple. *Spatial and logical arrangement of an example quadtree*. 2020. URL: https://developer.apple.com/documentation/gameplaykit/gkquadtree.

[Aro03] Sanjeev Arora. "Approximation schemes for NP-hard geometric optimization problems: A survey". In: *Mathematical Programming* 97.1-2 (2003), pp. 43–69.

[Aro98] Sanjeev Arora. "Polynomial time approximation schemes for Euclidean traveling salesman and other geometric problems". In: *Journal of the ACM (JACM)* 45.5 (1998), pp. 753–782.

[Ass83] Patrice Assouad. "Plongements lipschitziens dans $\mathbb{R}^n$". fr. In: *Bulletin de la Société Mathématique de France* 111 (1983), pp. 429–448. DOI: 10.24033/bsmf.1997. URL: http://www.numdam.org/item/BSMF_1983__111__429_0.

[BGK16] Yair Bartal, Lee-Ad Gottlieb, and Robert Krauthgamer. "The traveling salesman problem: low-dimensionality implies a polynomial time approximation scheme". In: *SIAM Journal on Computing* 45.4 (2016), pp. 1563–1581.

[Bie20] Therese Biedl. *CS 762: Graph-theoretic algorithms; Lecture notes of a graduate course University of Waterloo*. 2020.

[Bod+13] Hans L. Bodlaender et al. "Deterministic single exponential time algorithms for connectivity problems parameterized by treewidth". In: *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 7965 LNCS.PART 1 (2013), pp. 196–207. ISSN: 03029743. DOI: 10.1007/978-3-642-39206-1_17.

[CE12] Bruno Courcelle and Joost Engelfriet. *Graph structure and monadic second-order logic*. Cambridge University Press, 2012.

[Chr76] Nicos Christofides. *Worst-case analysis of a new heuristic for the travelling salesman problem*. Tech. rep. Carnegie-Mellon Univ Pittsburgh Pa Management Sciences Research Group, 1976.

[CI11] Chandra Chekuri and Sungjin Im. *CS 598CSC: Approximation Algorithms*. 2011.

[Cyg+11] Marek Cygan et al. "Solving connectivity problems parameterized by treewidth in single exponential time". In: *2011 IEEE 52nd Annual Symposium on Foundations of Computer Science*. IEEE. 2011, pp. 150–159.

[Der16] Sam Derbyshire. *File:Cantor set binary tree.svg*. Apr. 2016. URL: https://commons.wikimedia.org/wiki/File:Cantor_set_binary_tree.svg.

[Epp10] David Eppstein. *File:Tree decomposition.svg*. Mar. 2010. URL: https://commons.wikimedia.org/wiki/File:Tree_decomposition.svg.

[Esm06] Esmil. *File:Vector norms.svg*. Apr. 2006. URL: https://commons.wikimedia.org/wiki/File:Vector_norms.svg.

[Fak+03] Jittat Fakcharoenphol et al. "An improved approximation algorithm for the 0-extension problem." In: *SODA*. Vol. 3. Citeseer. 2003, pp. 257–265.

[Fel+18] Andreas Emil Feldmann et al. "A $(1+\varepsilon)$-Embedding of Low Highway Dimension Graphs into Bounded Treewidth Graphs". In: *SIAM Journal on Computing* 47.4 (2018), pp. 1667–1704.

[GKL03] Anupam Gupta, Robert Krauthgamer, and James R Lee. "Bounded geometries, fractals, and low-distortion embeddings". In: *Foundations of Computer Science* (2003).

[Gkn14] Gknor. *File:Kuratowski fan.svg*. Nov. 2014. URL: `https://commons.wikimedia.org/wiki/File:Kuratowski_fan.svg`.

[Jor14] Michael Jorgensen. "VOLUMES OF n-DIMENSIONAL SPHERES AND ELLIPSOIDS". In: (2014). URL: `https://www.whitman.edu/documents/Academics/Mathematics/2014/jorgenmd.pdf`.

[Kar] Asaf Karagila. *Is any compact metric totally disconnected space homeomorphic to a compact subspace of a Cantor space?* Mathematics Stack Exchange. URL:https://math.stackexchange.com/q/395935 (version: 2013-05-19). eprint: `https://math.stackexchange.com/q/395935`. URL: `https://math.stackexchange.com/q/395935`.

[Mat13] Jirí Matousek. *Lectures on Discrete Geometry*. Springer, 2013.

[NH02] T. S. E. Ng and Hui Zhang. "Predicting Internet network distance with coordinates-based approaches". In: *Proceedings.Twenty-First Annual Joint Conference of the IEEE Computer and Communications Societies*. Vol. 1. 2002, 170–179 vol.1.

[PW11] FengJi Peng and ShengYou Wen. "Fatness and thinness of uniform Cantor sets for doubling measures". In: *Science China Mathematics* 54.1 (2011), pp. 75–81.

[Rud+64] Walter Rudin et al. *Principles of mathematical analysis*. Vol. 3. McGraw-hill New York, 1964.

[SS10] Allan Scott and Ulrike Stege. "Parameterized pursuit-evasion games". In: *Theoretical computer science* 411.43 (2010), pp. 3845–3858.

[Tal04] Kunal Talwar. "Bypassing the embedding: algorithms for low dimensional metrics". In: *STOC '04: Proceedings of the thirty-sixth annual ACM symposium on Theory of computing*. Association for Computing Machinery, Inc., Jan. 2004, pp. 281–290. ISBN: 1-58113-852-0. URL: `https://www.microsoft.com/en-us/research/publication/bypassing-the-embedding-algorithms-for-low-dimensional-metrics/`.

[TSL00] Joshua B. Tenenbaum, Vin de Silva, and John C. Langford. "A Global Geometric Framework for Nonlinear Dimensionality Reduction". In: *Science* 290.5500 (2000), pp. 2319–2323. ISSN: 0036-8075. DOI: `10.1126/science.290.5500.2319`. eprint: `https://science.sciencemag.org/content/290/5500/2319.full.pdf`. URL: `https://science.sciencemag.org/content/290/5500/2319`.

[WS11] David P. Williamson and David B. Shmoys. *The Design of approximation algorithms*. Cambridge University Press, 2011.

[WWW15] Chun Wei, Shengyou Wen, and Zhixiong Wen. "Doubling measures on uniform Cantor sets". In: *Journal of Mathematical Analysis and Applications* 430.1 (2015), pp. 500–516.

8 Appendix

List of Figures

1 The unit balls in $\mathbb{R}^2$ corresponding to the ℓ_1-norm, Euclidean distance, and ℓ_∞-norm, courtesy of [Esm06]. 4
2 A greedily constructed δ-net. The solid points have been chosen as net points. Notice how choosing the red point would have covered both its neighbours so this net is not minimal. . . 5
3 A graph and an accompanying tree decomposition of width 2, courtesy of [Epp10]. 8

4 A minimum vertex cover in a tree. 9
5 Points on the plane and a corresponding quadtree, courtesy of [App20]. 10
6 The hierarchical partition of a metric space labelled according to a random permutation π, courtesy of [Tal04]. 12
7 The accompanying split-tree to Figure 6 labelled with the balls using which clusters were "cut". 13
8 A uv-path of portals in the embedding obtained from the split-tree in Figure 7. 17
9 The MST based 2-approximation of metric TSP, courtesy of [CI11]. 19
10 Obtaining W from S at a single point $u \in Y$. 22
11 The Cantor set and corresponding "parse-tree", courtesy of [Der16]. 24
12 The Knaster-Kuratowski Fan is a connected metric space which becomes totally disconnected from the removal of a single point, courtesy of [Gkn14]. 25